

An *explicit* geometric description of the polytope $P = Q^\Delta$ is given in Exercise 6.27.

(d) Polytopes Violating the Isotopy Conjecture

Recall from Section 4.4 that the “realization space” of a polytope is an elementary semialgebraic set, and that elementary semialgebraic sets can be arbitrarily complicated spaces: they can be disconnected, with holes, and so forth (Exercise 4.22).

However, Steinitz’ Theorem 4.11 states that for every 3-dimensional polytope the realization space $\mathcal{R}(P)$ is contractible, and thus connected. (To get this right, we had to fix an affine basis in our Definition 4.10 of realization spaces, to make sure that the “reflection” doesn’t create a second component of the realization space.)

In other words, any two 3-polytopes of the same combinatorial type and orientation can always be continuously deformed into each other, such that each intermediate object is a 3-polytope of the same combinatorial type. The same is not hard to show for d -polytopes with at most $d + 3$ vertices, using Gale diagrams.

However, this “isotopy property” fails even for 4-polytopes: Kleinschmidt constructed a 4-dimensional example with 10 vertices [114]; its combinatorial type is obtained by glueing two copies of the 8-vertex Kleinschmidt polytope of Exercise 6.18 in their octahedron facets in an “incompatible way.” See Mněv [408, p. 530] and Bokowski & Guedes de Oliveira [115]. A systematic construction method for 4-dimensional counterexamples is provided by Richter-Gebert’s Universality Theorem for 4-polytopes [459].

Here we start a construction with a planar point configuration, where the *isotopy conjecture* [463] [234] fails — and transfer this result to polytopes.

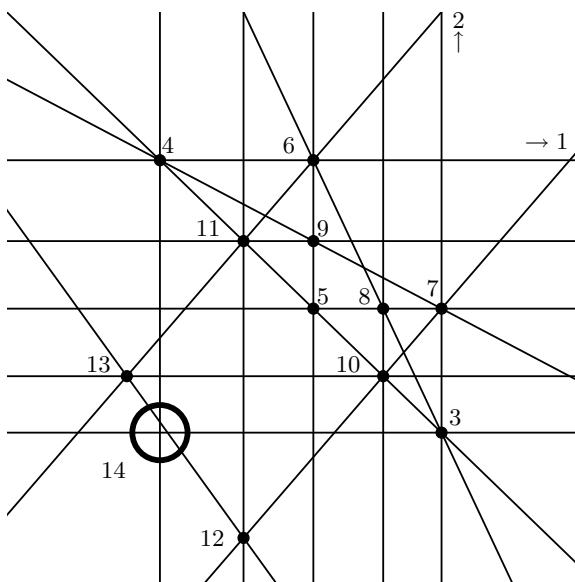
Example 6.24 (A nonisotopic 24-polytope with 28 vertices).

The “isotopy property” fails for d -polytopes with $d + 4$ vertices: for that, we can start from a planar point configuration that does not satisfy isotopy. The smallest nonisotopic planar point configurations that are currently known have 14 points. The first such configuration was found by Suvorov [537] [96, p. 363]. Here we present the nicest and newest one, found by Richter-Gebert [458].

For this we refer to the figure on the next page: it represents Richter-Gebert’s example, with two points drawn “at infinity” (which we interpret as “very far away,” to get an affine configuration).

The key property of the figure can be seen as follows. Try to construct a new configuration of 14 points in the (projective) plane, with the condition that point sets that are collinear in the old figure also have to be collinear in the new figure. After a projective transformation, we may assume that the points 1, 2, 3, and 4 (a “projective basis”) are positioned as in the old figure.

Then we choose the point 5 on the diagonal through 3 and 4. Thus we have five points which then successively determine the points 6, 7, ..., 13. The fourteenth point 14 is then placed at the intersection of the lines $\overline{1,3}$ and $\overline{2,4}$, but not on the line $\overline{12,13}$: this is possible if the point 5 had been drawn slightly Southeast of the center, as in our figure, or if it is taken slightly Northwest of the center, which results in a figure that is a reflection of our figure; however, it is impossible (it results in a different configuration, with 12, 13, 14 collinear) to get a realization of the same configuration which is itself symmetric with respect to the $x = y$ diagonal: for that we'd have to choose 5 in the center, and would get 12, 13 and 14 collinear.



A configuration with this effect is not too hard to construct, but this example has a stronger property: If the point 5 is chosen close enough to the midpoint of the segment $[3, 4]$, then the “Southeast” and “Northwest” realizations not only have the same collinearities (and thus the same unsigned cocircuits, the same matroid), but they have the same *signed cocircuits* — both realizations yield the same *oriented matroid*. We do not know whether one really needs 14 points for this effect: can you do with less (Problem 6.26*)?

From this, one can easily construct an affine planar Gale diagram that has a disconnected realization space. However, we must also make sure that every polytope that is (only!) combinatorially equivalent to the one we construct has the same diagram. A very “aggressive” method is to replace every point of the diagram by a pair of positive and negative points: this yields the affine Gale diagram of a 24-polytope with only 28 vertices that has two isotopy classes of realizations. Implicitly, this is what the “Lawrence construction” does, which we will discuss below.

Another nonisotopic point configuration, of 21 points, is explicitly constructed in White [559]. A general construction method is due to Mnëv [408]. In particular, Mnëv’s universality theorem (see below) shows that the “realization space” for planar point configuration can be arbitrarily complicated.

Suvorov [537] and Jaggi et al. [290] furthermore construct non-isotopic configurations whose points are in general position. From these, one can — using a technique of Sturmfels [117, Thm. 6.5] — get *simplicial* polytopes that violate the isotopy conjecture. This yields examples for a much more general “universality theorem for polytopes,” described in the next section.

6.6 Rigidity and Universality

We have characterized Gale diagrams of polytopes in Corollary 6.20. To make statements about polytopes of a fixed combinatorial type, however, it is not sufficient to look at a specific Gale diagram: we have to make sure that the statement we make holds for *all polytopes combinatorially equivalent to the given polytope*.

Here we have to deal with two different notions of equivalence. We have noted that two polytopes may be combinatorially equivalent but have different oriented matroids. (See, for example, the octahedra of Example 6.3.) However, any two polytopes with the same oriented matroid are combinatorially equivalent: they have the same covectors, hence the same positive covectors, and hence the same cofaces, and thus the same faces.

To make Gale diagrams useful for high-dimensional polytopes, we have to get a hold on all Gale diagrams representing a combinatorial equivalence class of polytopes. This is not simple, and it would lead into a discussion of “partial oriented matroids” that we want to avoid (there is not much theory for that, either). Instead, we restrict ourselves to an important special case: when only one oriented matroid is possible for a given convex polytope.

Definition 6.25. The oriented matroid of a d -polytope $P \subseteq \mathbb{R}^d$ is *rigid* if every polytope P' that is combinatorially equivalent to P has the same oriented matroid.

Here our convention is to identify the vertex sets of P and of P' with $[n]$ in a way that is compatible with the combinatorial equivalence. With this, *combinatorial equivalence* means that the oriented matroids $\mathcal{M}(P)$ and $\mathcal{M}(P')$ have the same set of *positive cocircuits*, and *rigidity* means that this implies that *all* cocircuits of P and of P' coincide:

$$\mathcal{C}^*(P) \cap \{+, 0\}^n = \mathcal{C}^*(P') \cap \{+, 0\}^n \implies \mathcal{C}^*(P) = \mathcal{C}^*(P').$$

For example, triangular prisms are rigid, but octahedra are not. The concept of rigidity is only interesting because there are some rigid polytopes around, although “most” polytopes are not rigid. Here is one construction to get rigid polytopes.